Tokenomics and the AI Factory Stack
Why Asia Needs More Datacenters

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The AI Token Factory Economics Stack

	The AI Token Factory Economics Stack				
	Layer	**Units**	**H200**	**Analysis Methodology**	**SemiAnalysis Team**
Datacenters	Colocation Cost	*USD/kW/mth*	$130	DC Market Surveys	AI Datacenters
	Electricity Cost	*USD/kWh*	$0.087	Power Grid/Market Statistics	
	Capital Cost per GPU	***USD/GPU***	**$36,620**	Semicon Manufacturing Analysis	IC Design, Networking, Systems
Neoclouds	Cost of Capital	*%, per annum*	13.3%	Neocloud Market Surveys	AI Neoclouds
	Total Cost of Ownership	***USD/hr/GPU***	**$1.75**		
	Neocloud Pre-tax Profit Margin	*%*	12.7%		
	Neocloud Rental Price	***USD/hr/GPU***	**$2.00**		
Inference Providers	Inference Throughput/GPU[1]	*Tokens/s/GPU*	313	Llama 405B FP8 Benchmarking Runs	AI Engineering
	GPU Rental Cost	*USD/hr/GPU*	$2.00	Neocloud Market Surveys	
	Effective Utilization Rate	*%*	75%	Systems Analysis	
	Inference Compute Cost	***USD/M Tok***	**$2.37**		
	Inference Provider Gross Margin	*%*	13.9%		
	Token Selling Price	***USD/M Tok***	**$2.75**		
Applications	Tokens/User/Month	*Tok/user/mth*	6,000,000	Inference market research, surveys, proprietary dataset analysis	AI Tokenomics
	Token Unit Cost	*USD/M Tok*	*$2.75*		
	Monthly Active Users	*M Users*	*1.8*		
	Average Revenue per User	*USD/mth*	$25.00		
	Gross profit per User/Mth	***USD/mth***	**$8.50**		
	Application Gross Margin	*%*	34.0%		

1. Llama 405B FP8 on H200 using VLLM. Uses 1k input, 4k output tokens, ~190 seconds end to end latency.

Source: SemiAnalysis estimates.

Total Cost of Ownership Summary for H200

TCO Summary for H200		
	Unit	H200 SXM
Cluster Capital Costs		
GPU Provider Content (per GPU)	USD	[redacted]
GPU Provider Content + Warranty (per Server)	USD	[redacted]
Other Server Costs	USD	[redacted]
Server Cost	USD	$212,579
Service, Networking, Storage, Software, Others	USD	$80,377
Total Upfront Cluster Capex, per Server	**USD**	**$292,956**
Total Upfront Cluster Capex, per Logical GPU	**USD**	**$36,620**
Weighted Average Cost of Capital	%	13.3%
Useful Life in Years	Years	4
Total Cluster Capital Costs per Month per Server	**USD/mth**	**$7,896**
Capital Cost per Unit, per Hour	USD/hr/GPU	$1.35
Operating Cost per Unit, per Hour	USD/hr/GPU	$0.39
Total Cost per Unit per Hour	**USD/hr/GPU**	**$1.75**

Source: SemiAnalysis estimates.

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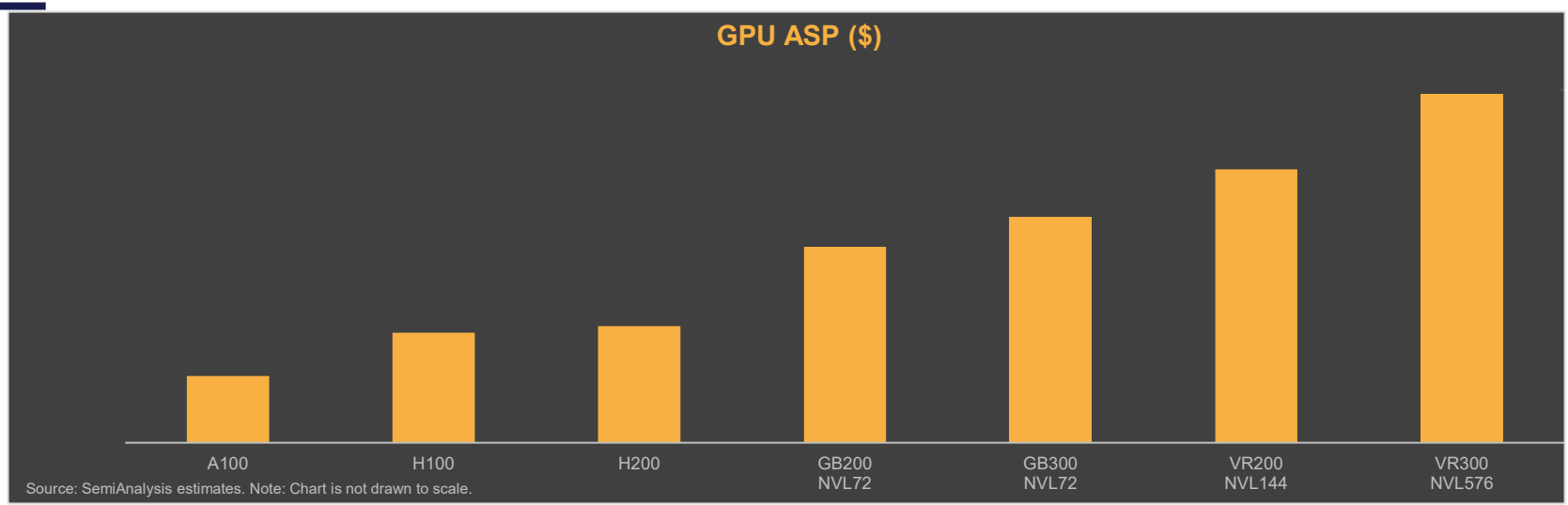
Accelerator Roadmap – GPU Content Increases



	A100	H100	H200	GB200 NVL72	GB300 NVL72	VR200 NVL144	VR300 NVL576
Memory Capacity (GB)	80	80	141	192	288	288	1,024
Memory BW (TB/s)	2.0	3.4	4.8	8.0	8.0	13.0	32.0
Logic Die Configuration	1 x Reticle Sized GPU			2 x Reticle Sized GPU		2 x Reticle GPU, 2 x I/O	4 x Reticle GPU, 4 x I/O
Interposer Size (# Reticles)	1.9x	1.9x	1.9x	3.3x	3.3x	4.8x	9.6x
TDP (W)	400	700	700	1,200	1,400	1,800	2,800
Dense Flops FP8 (TFLOPs)	312	1,979	1,979	5,000	5,000	16,667	33,333

Source: SemiAnalysis estimates.

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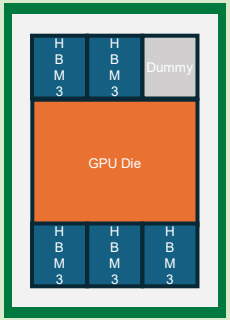
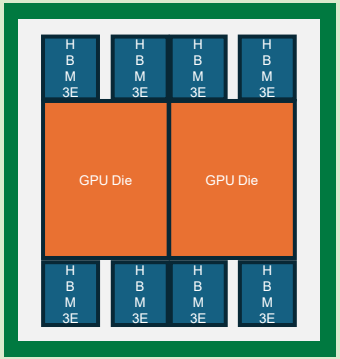
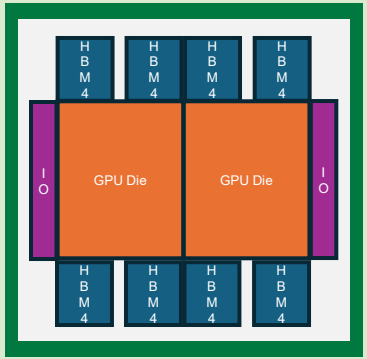
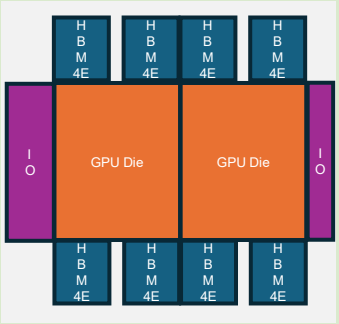
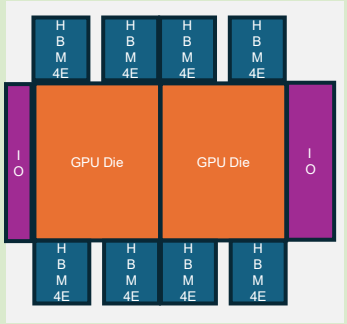
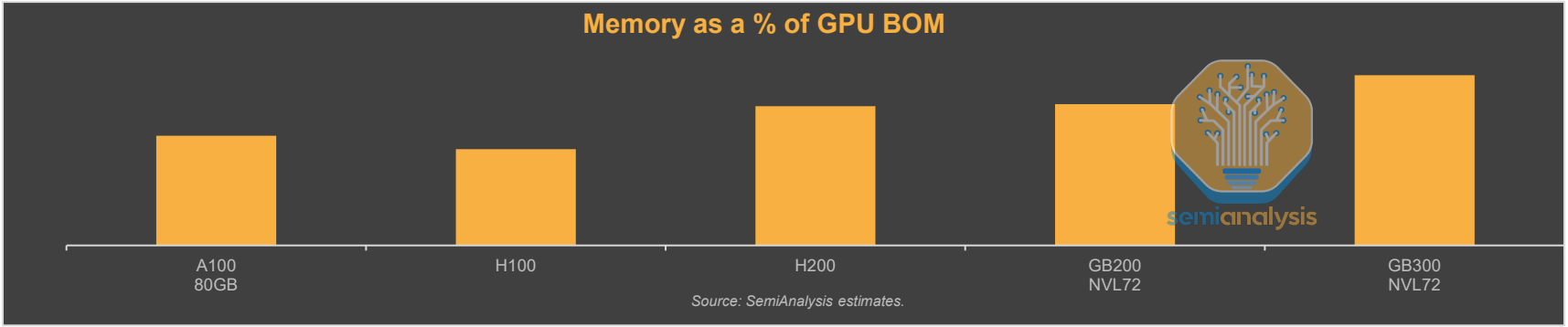
Evolution of NVIDIA GPU Packages

Hopper	Blackwell	Rubin	Rubin Ultra
• 1 x Reticle Sized GPU • 5 x HBM3 and 1 x Dummy Die	• 2 x Reticle Sized GPU • 8 x HBM3E	• 2 x Reticle Sized GPU • 8 x HBM4 • 2 x I/O Chiplet	• 4 x Reticle Sized GPU • 16 x HBM4E • 4 x I/O Chiplet
			 

Source: SemiAnalysis.

HBM Overview – NVIDIA Roadmap



NVIDIA HBM Roadmap

	A100 80GB	H100	H200	GB200 NVL72	GB300 NVL72	VR200 NVL144	VR300 NVL576
HBM Type	2E	3	3E	3E	3E	4-24Gb	4E
Gb per Layer (Gb)(1)	16	16	24	24	24	24	32
Layers per Stack (#)	8	8	8	8	12	12	16
GB per Stack (GB)	16	16	24	24	36	36	64
HBM Stacks (#)	5	5	6	8	8	8	16
Total Capacity (GB)	**80**	**80**	**144**	**192**	**288**	**288**	**1,024**
Increase vs A100	**1.0x**	**1.0x**	**1.8x**	**2.4x**	**3.6x**	**3.6x**	**12.8x**
Memory BW (TB/s)	**2.0**	**3.4**	**4.8**	**8.0**	**8.0**	**13.0**	**32.0**
Increase vs A100	**1.0x**	**1.6x**	**2.4x**	**3.9x**	**3.9x**	**6.4x**	**15.7x**

Note: (1) 8Gb = 1GB.

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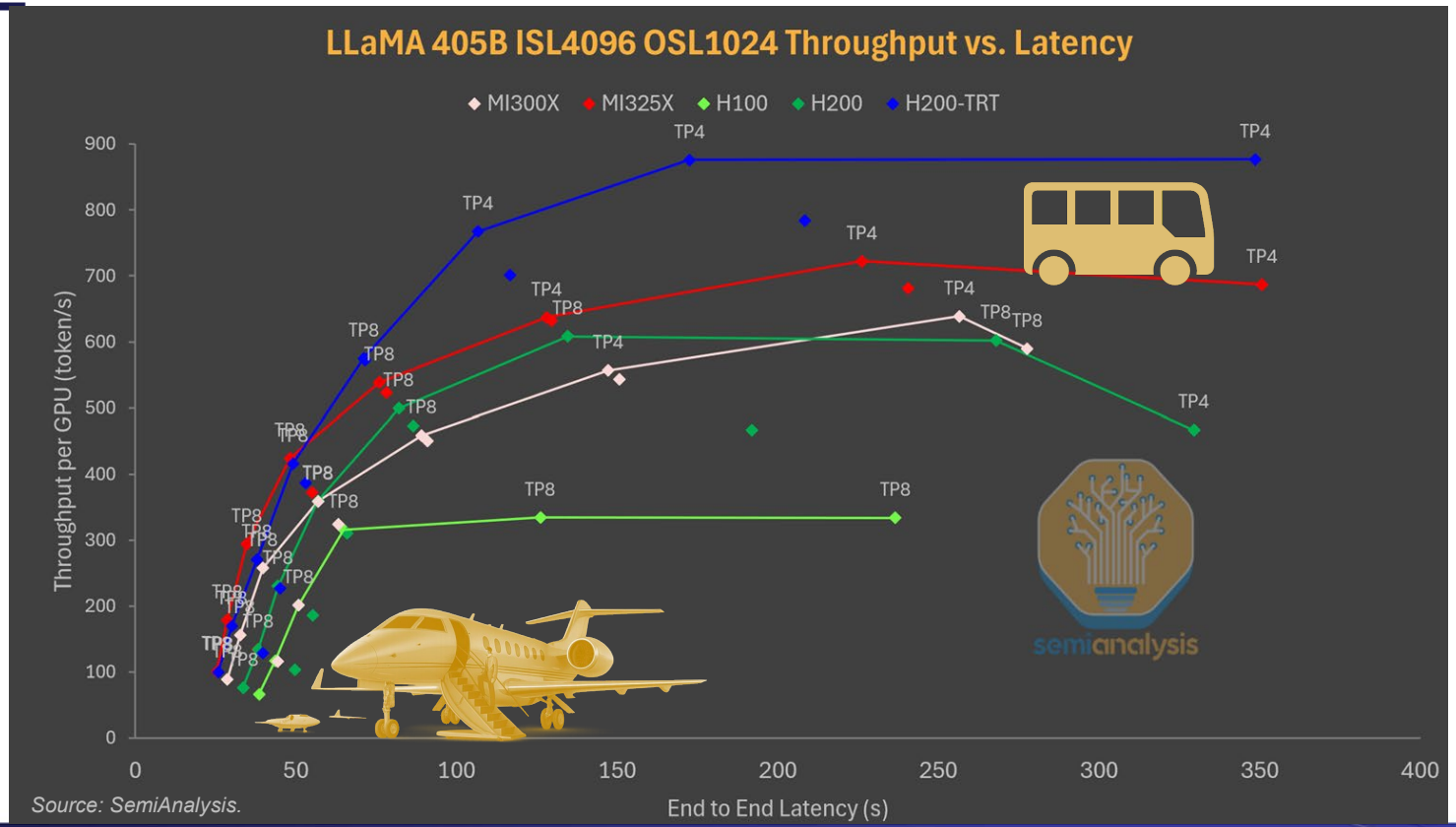
AI Triad Tradeoffs: Latency vs Throughput



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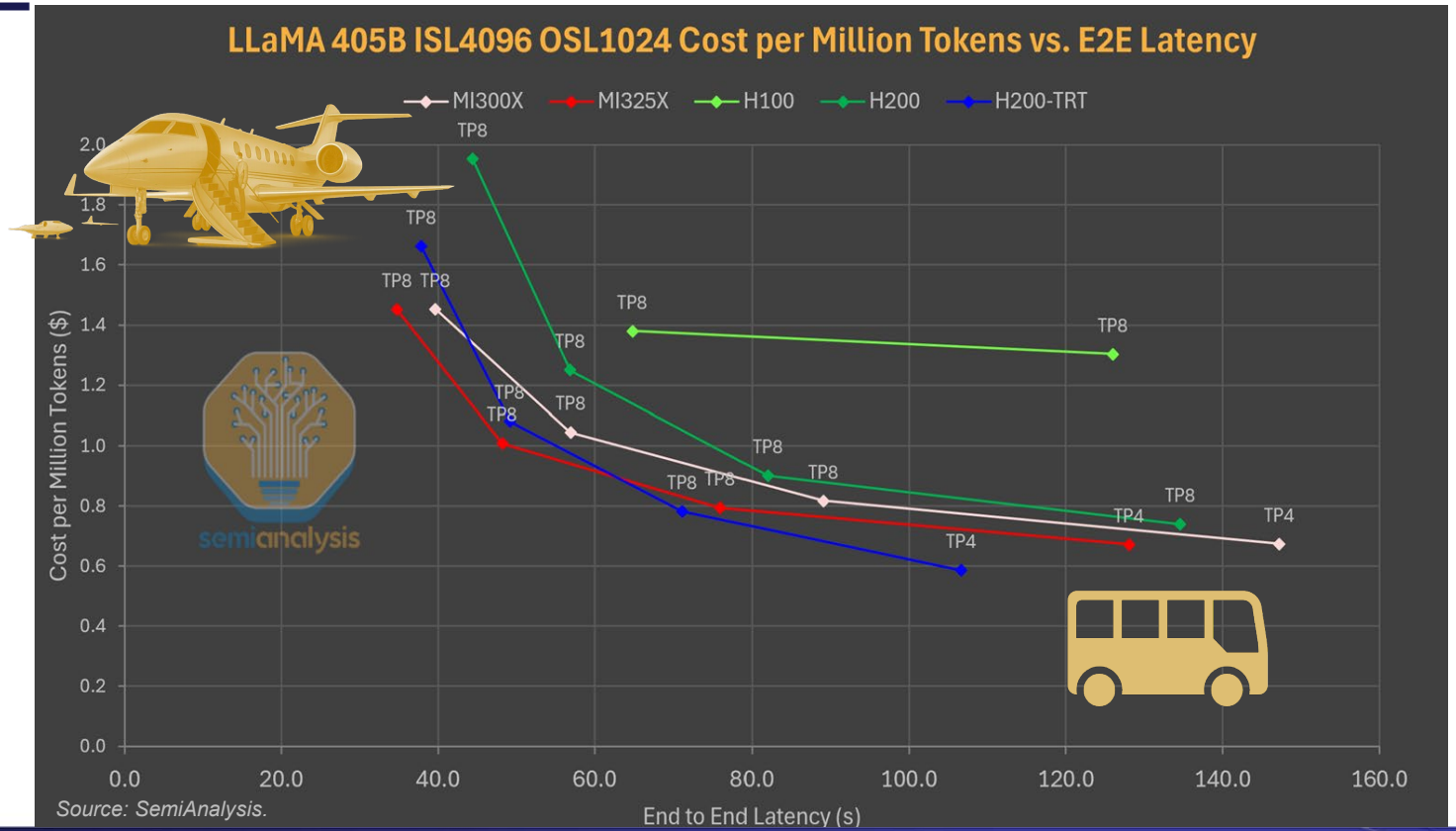
AI Triad Tradeoffs: Latency vs Cost



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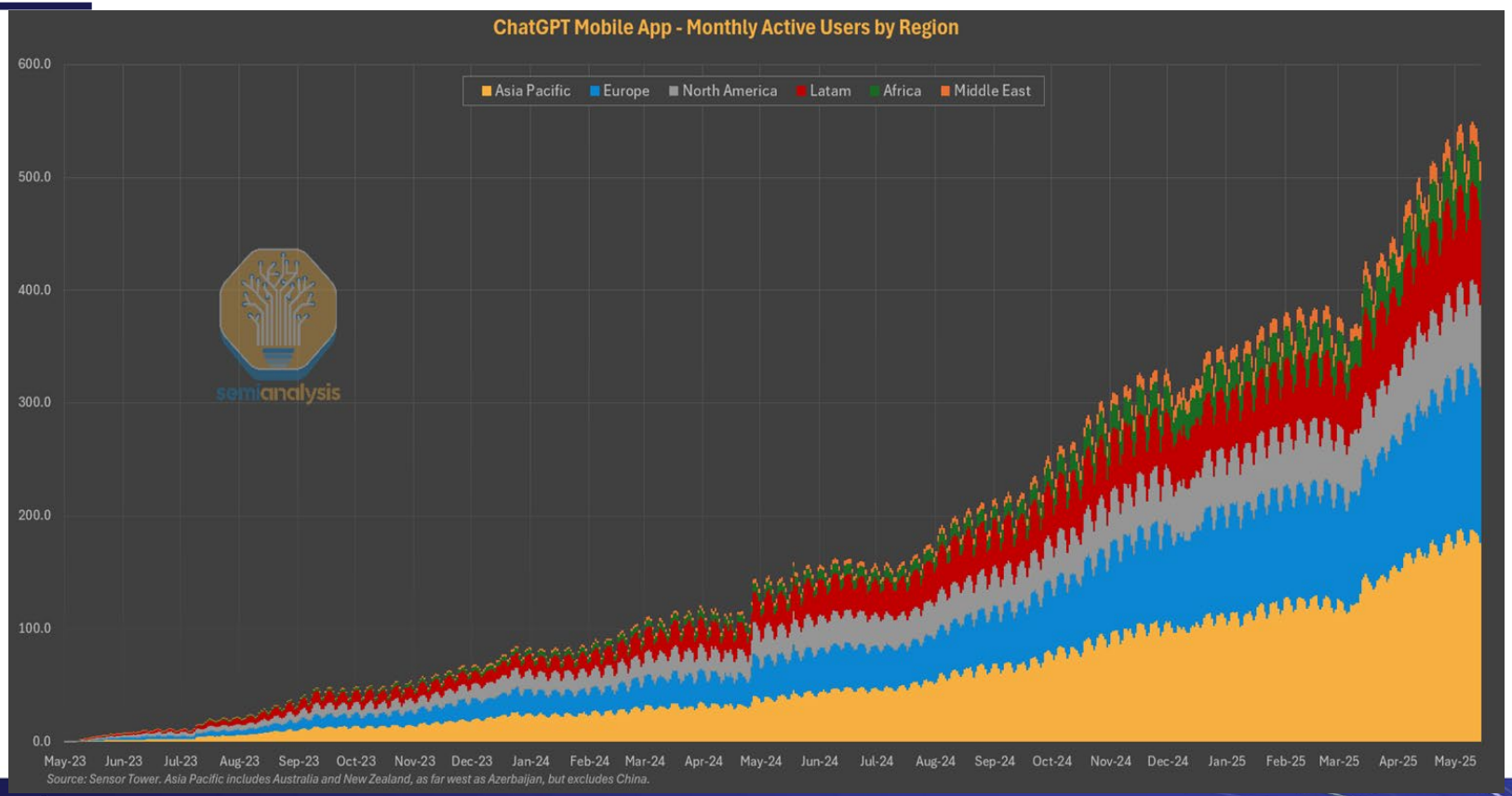
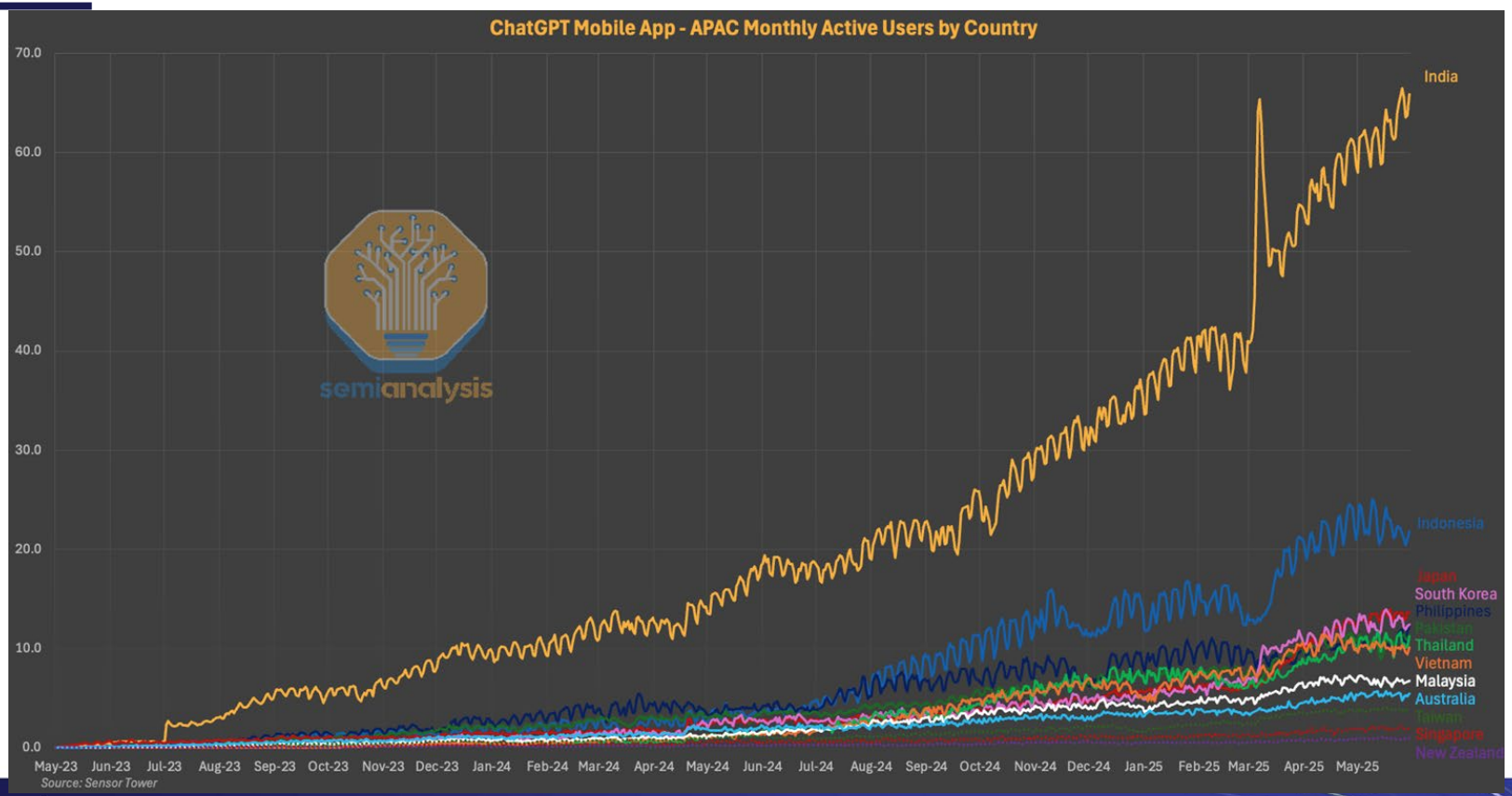
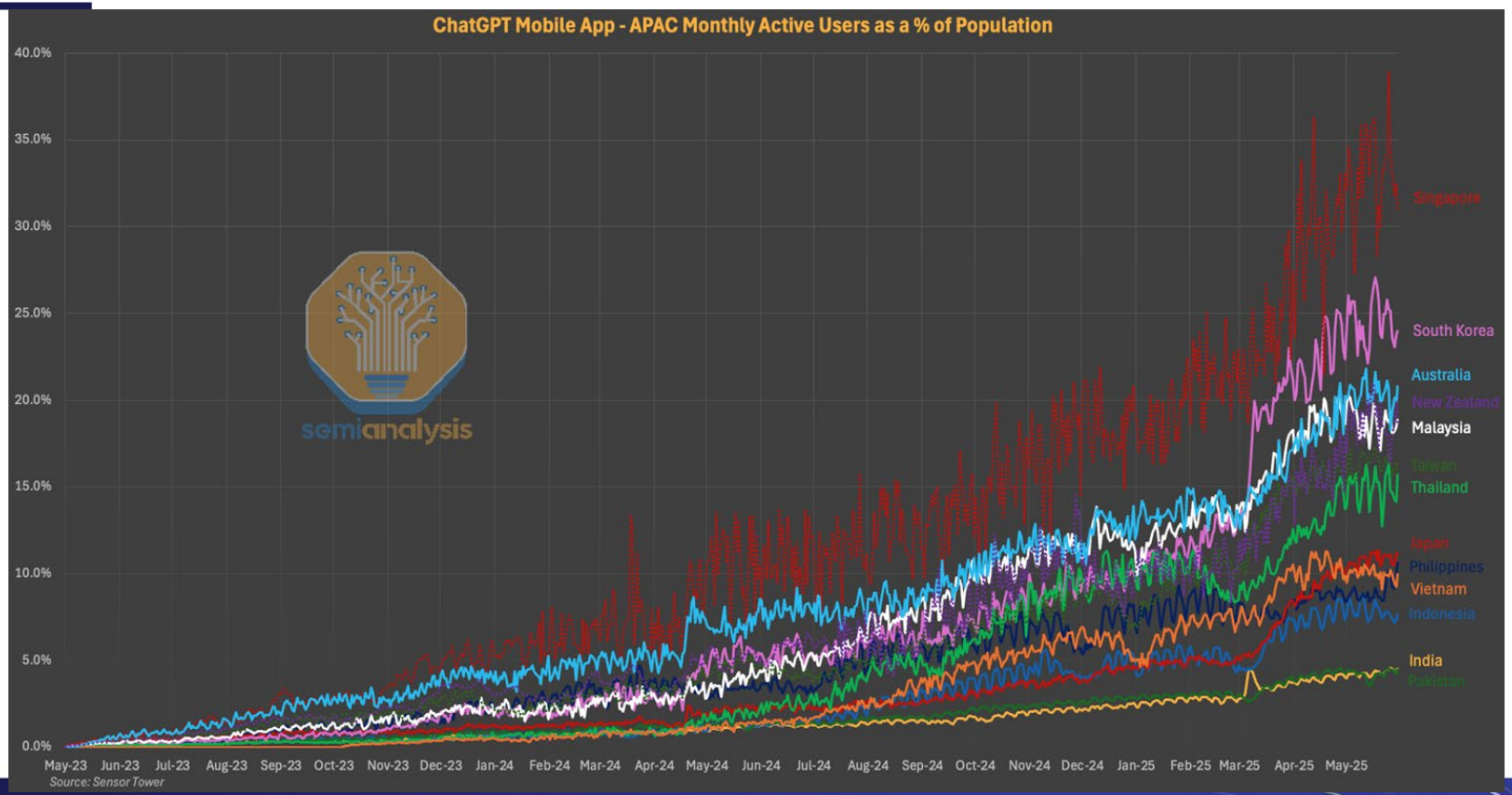
Asia Pacific is the largest region for ChatGPT



India has the largest MAU in Asia for ChatGPT…



... But it has among the lowest user penetration rates in the region!



Inference Demand Analysis Framework: Key Ingredients

- Population
- Monthly Average User (MAU) Penetration Rate
- Daily Active (DAU) to Monthly Active (MAU) User Ratio
- Queries per Day
- Tokens Per Query
- Inference System Throughput
- All-in Critical IT Power per GPU

Inference Tokenomics - Asia Pacific Consumer in May 2025				
Measure	**Unit**	**ChatGPT**	**The Internet**	**Analysis Method**
Asia Pacific Population[1]	M	3,370	3,370	App and Website Engagement data
MAU Penetration Rate	%	6.3%	66.0%	
Monthly Active Users (MAU)[2]	**M**	**211**	**2,224**	
DAU to MAU Ratio	%	40.4%	80.0%	
Daily Active Users (DAU)	**M**	**85**	**1,779**	
Queries Per Day	Queries/Day	7.5		Inference Provider Survey
Tokens per Query	Tok/Query	3,000		
Tokens Per Day	**Trillion Tok/Day**	**1.9**		
Inference Throughput/GPU[3]	Tokens/s/GPU	313		Workload Benchmarking
GPUs Required	GPUs	70,990		
All-in Critical IT Power per GPU	W	1,302		AI Total Cost of Ownership Analysis
AI Datacenter Capacity	**MW**	**92.4**		

1. Asia Pacific includes Australia and New Zealand, as far west as Azerbaijan, but excludes China.
2. Estimated de-duplicated combined mobile app + web MAU, approximately 17% desktop + mobile web and 83% mobile app.
3. Llama 405B FP8 on H200 using VLLM. Uses 1k input, 4k output tokens, ~190 seconds end to end latency.

Source: SemiAnalysis estimates.

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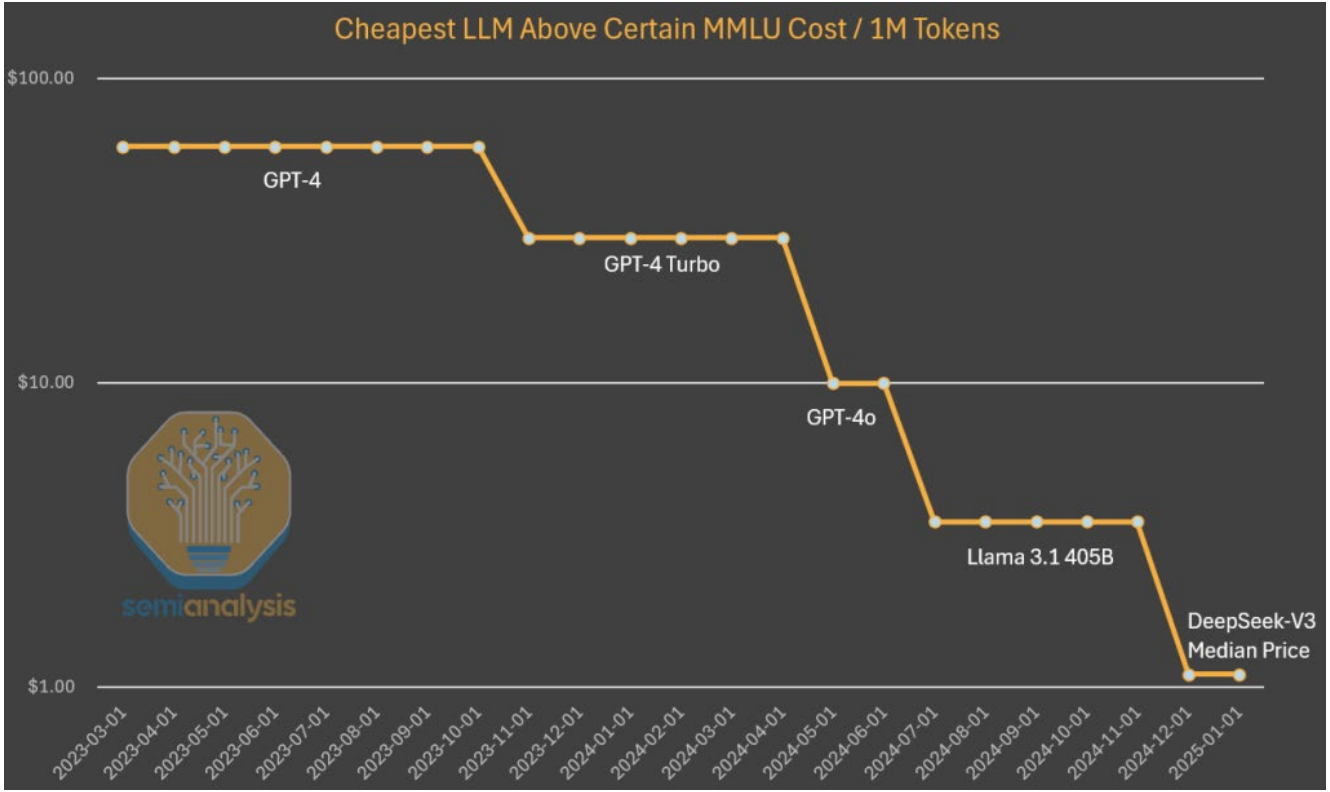
Token Demand Debates: Trends in Cost per Token



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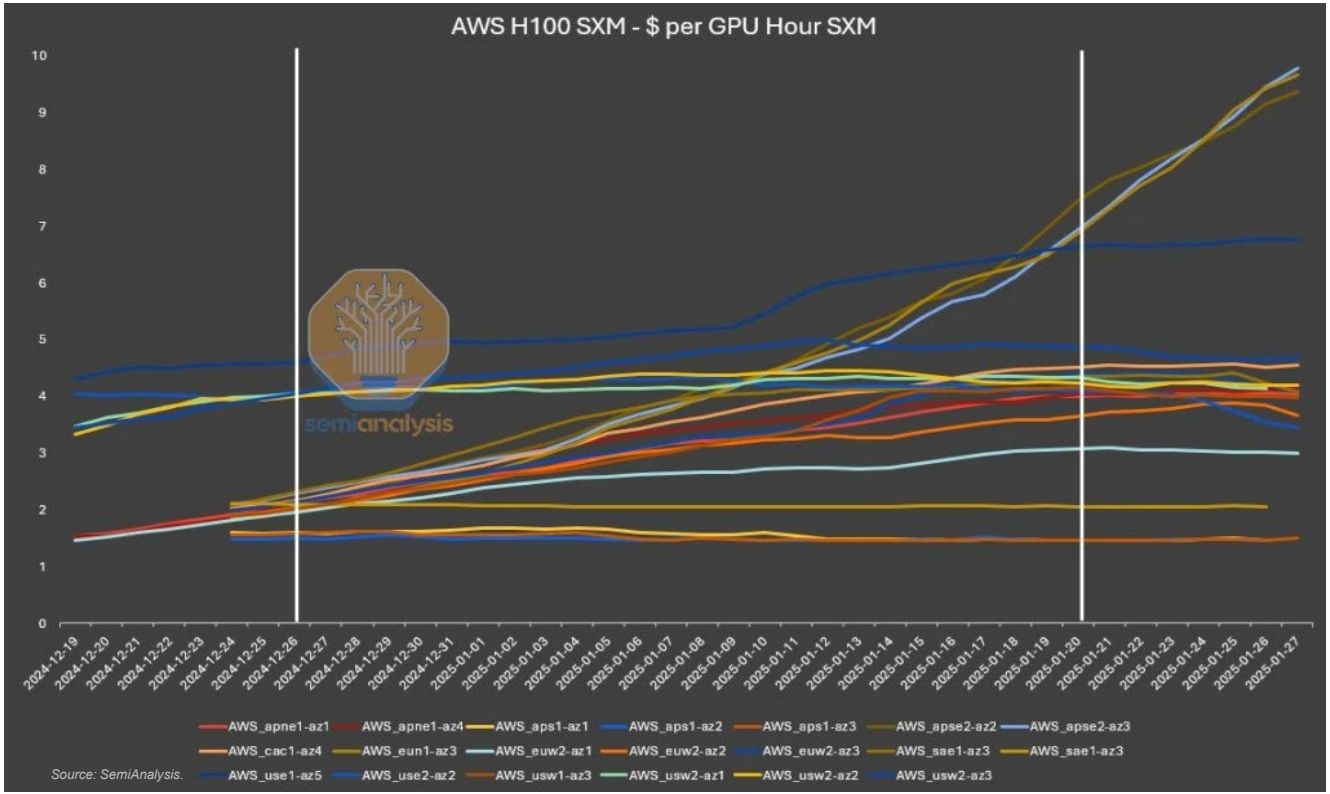
Token Demand Debates: Jevons' Paradox at Work!



Inference Demand Analysis – Broad Estimates

Inference Tokenomics - Asia Pacific Consumer in May 2025: ChatGPT						
Measure	**Unit**	**2025**	**2028**	**2030**	**The Internet**	**Analysis Method**
Asia Pacific Population[1]	M	3,370	3,457	3,516	3,370	App and Website Engagement data
MAU Penetration Rate	%	6.3%	23.7%	46.7%	66.0%	
Monthly Active Users (MAU)[2]	**M**	**211**	**820**	**1,640**	**2,224**	
DAU to MAU Ratio	%	40.4%	50.9%	66.1%	80.0%	
Daily Active Users (DAU)	**M**	**85**	**417**	**1,085**	**1,779**	
Queries Per Day	Queries/Day	7.5	25.2	44.2		Inference Provider Survey
Tokens per Query	Tok/Query	3,000	18,295	51,922		
Tokens Per Day	**Trillion Tok/Day**	**1.9**	**192.2**	**2,487.9**		
Inference Throughput/GPU[3]	Tokens/s/GPU	313	1,133	5,074		Workload Benchmarking
GPUs Required	GPUs	70,990	1,962,918	5,674,532		
All-in Critical IT Power per GPU	W	1,302	2,537	2,959		AI Total Cost of Ownership Analysis
AI Datacenter Capacity	**MW**	**92**	**4,980**	**16,791**		

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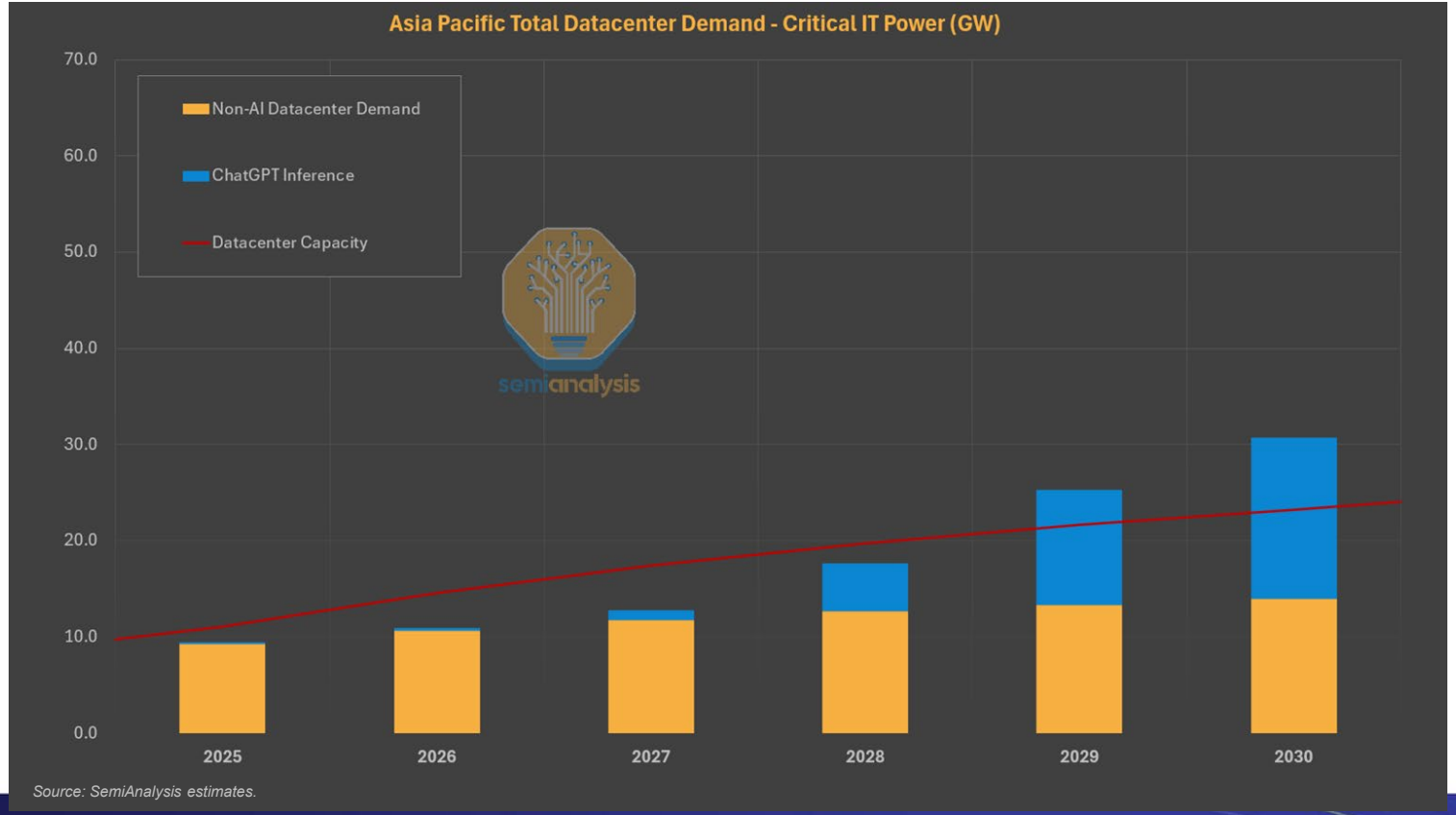

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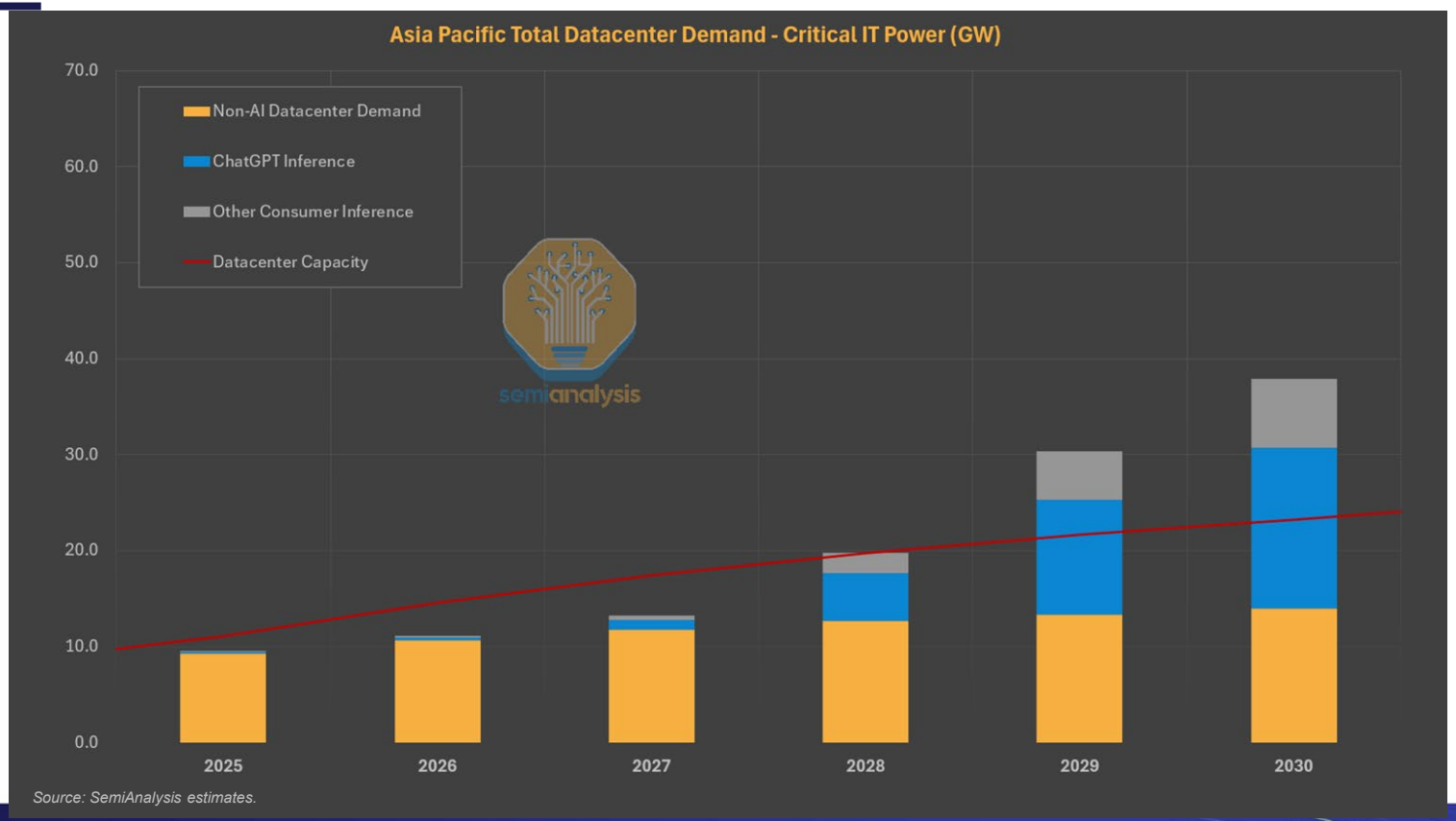


AI in Action: Consumers, Enterprises and Government

Consumer

Personal Productivity

ChatGPT
Gemini
deepseek

Digital Assistants

alexa
Google Assistant

Personalized Content & Recommendations

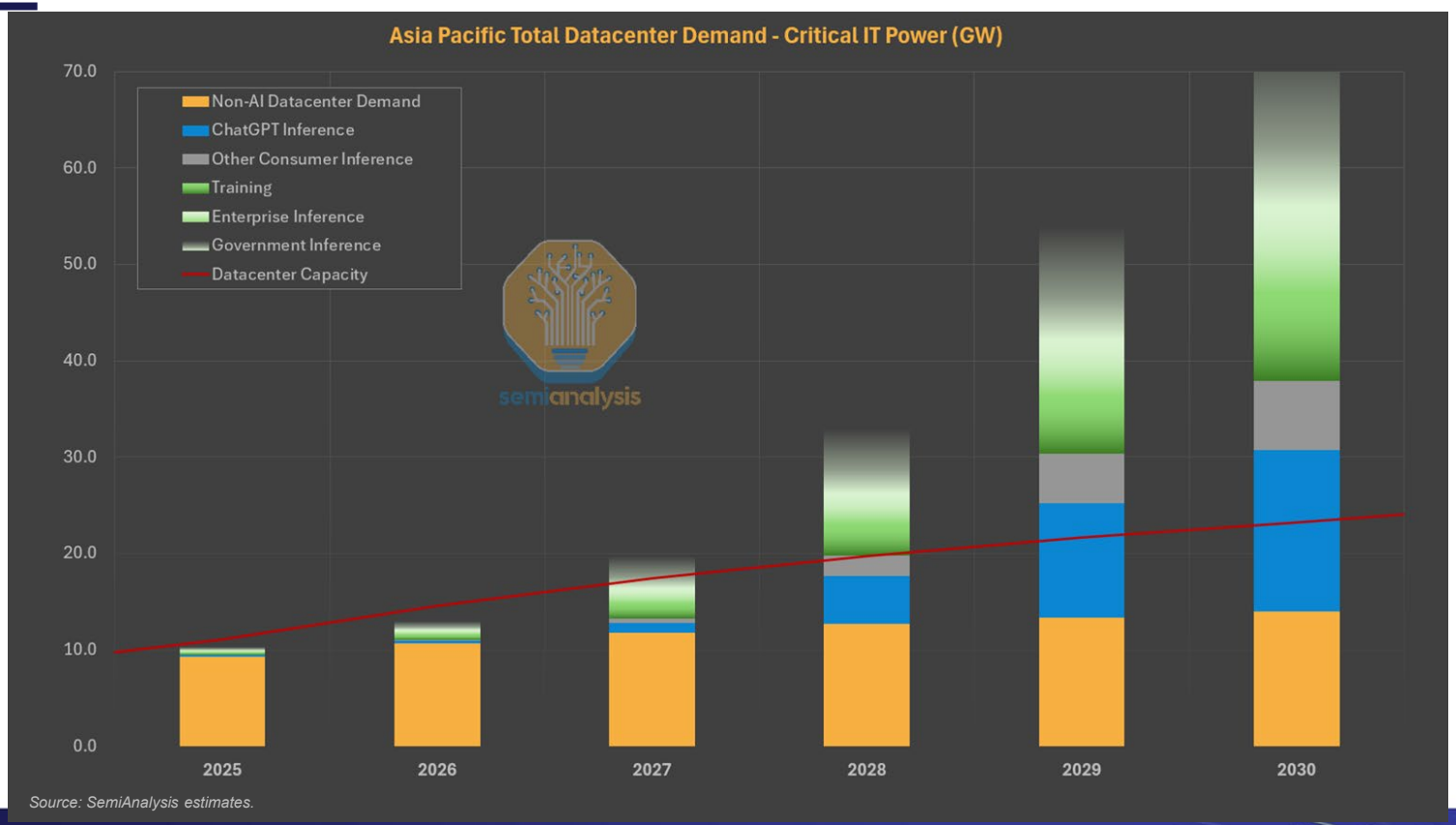
Enterprise

- Gen AI for Knowledge Work
- Manufacturing
- Healthcare
- Banking & Finance
- Supply-Chain & Logistics

Government

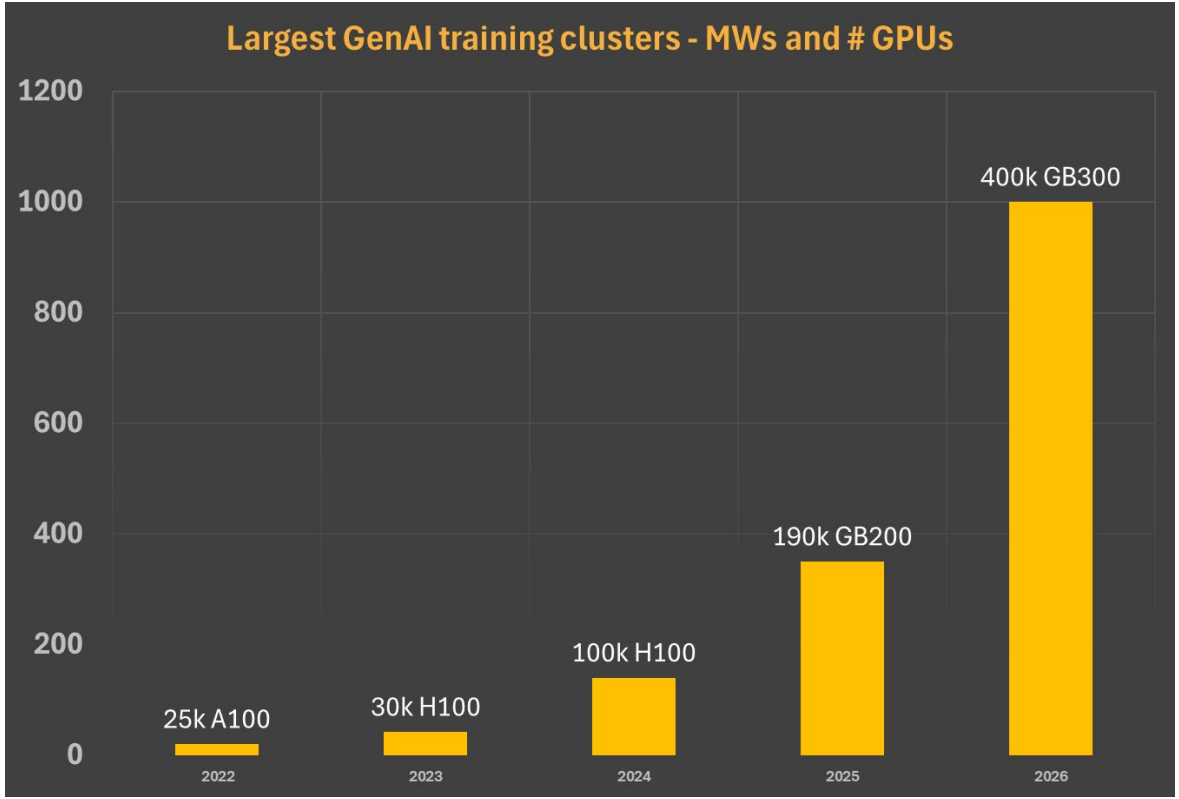
- Virtual Government Chatbot
- Defence & National Security
- Smart Nation Initiatives
- Policy Planning

Inference Demand Analysis



Datacenter architecture – AI training clusters

- Largest Training cluster sizes are booming
- 1GW training clusters are already under construction



Source: SemiAnalysis estimates.

AI Datacenter Clusters in the US – 220 MW Pictured…

Source: SemiAnalysis.

Stargate – 2GW by 2028

Source: SemiAnalysis.

Thank You!

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